

Operator-Resolved QFI Response and Matched-Filter Computation of $\eta_{\leq k}$

QFI chain notes

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1 Physics Question

The sharp question is not whether a single scalar “detects entanglement.” The sharp question is:

Given a state $|\psi\rangle$ and a generator G , through which observable algebra is the QFI response $|\delta G\rangle = (G - \langle G \rangle)|\psi\rangle$ visible?

The total response strength is measured by

$$F_Q = 4 \langle \delta G | \delta G \rangle .$$

The matched-filter ratio

$$\eta_{\mathcal{V}}^* = \frac{\|P_{\mathcal{V}|\psi}\|\delta G\|^2}{\|\delta G\|^2}$$

then asks what fraction of that response can be read out by observables inside a specified real Hermitian algebra $\mathcal{V}$.

Thus $\eta_{\mathcal{V}}^*$ is best understood as an *operator-resolved visibility of the QFI response*. It is not a universal entanglement monotone and should not be interpreted as a sure-fire diagnostic of long-range entanglement by itself. The choice of $\mathcal{V}$ is part of the physical question.

Example: Haldane-chain warning. In a spin-1 Haldane phase, local two-point correlations are short-ranged, but the entanglement/correlation structure is naturally exposed by nonlocal string operators. If one restricts $\mathcal{V}$ to ordinary local windows, $\eta_{\mathcal{V}}^*$ may remain small until large windows are included. That does *not* mean the Haldane phase is long-range entangled in the GHZ sense. It means the response to the chosen G is not efficiently visible in the local-window algebra. If $\mathcal{V}$ is enlarged to include the appropriate string algebra, the same short-range entangled SPT state can have a large $\eta_{\mathcal{V}}^*$.

This is the correct interpretation:

$\eta_{\mathcal{V}}^*$ measures visibility through $\mathcal{V}$, not entanglement itself.

Different choices of $\mathcal{V}$ expose different channels:

- local-window algebras diagnose local recoverability of the response;
- string algebras diagnose hidden or SPT response channels;
- scale-dependent families diagnose critical delocalization of the response.

2 Numerical Purpose

The important numerical observables for the corrected period-4 J_1 - J_2 - J_3 chain are

$$F_Q[\psi; G] \quad \text{and} \quad \eta_{\leq k}^{\star, \text{conn}} = \frac{F_{\text{EP}}[O_{\leq k}^{\star}]}{F_Q}, \quad (1)$$

where $G = S_{\pi}^z$ is the staggered magnetization and $O_{\leq k}^{\star}$ is the optimal matched-filter observable inside the real Hermitian subspace of connected Pauli windows of size at most k . For this specific paper figure, $\mathcal{V}$ is the connected-window algebra, so $\eta_{\leq k}$ means “local-window visibility of the staggered QFI response.” For the present path scan we need $k = 2, 4$ at

$$N = 16, 32, 64$$

along

$$\begin{aligned} D \rightarrow C : \quad & (J_1, J_2, J_3) = (1, t, 0), \\ C \rightarrow U : \quad & (J_1, J_2, J_3) = (1, 1, t), \\ U \rightarrow D : \quad & (J_1, J_2, J_3) = (1, t, t). \end{aligned}$$

Here J_1, J_2, J_3 label the period-4 nearest-neighbor bond pattern

$$J_1, J_2, J_1, J_3, J_1, J_2, J_1, J_3, \dots,$$

not first-, second-, and third-neighbor couplings.

3 Definitions

Let $|\psi\rangle$ be a normalized DMRG ground state. The generator is

$$G = S_{\pi}^z = \sum_{j=1}^N (-1)^j S_j^z. \quad (2)$$

The QFI response vector is

$$|\delta G\rangle = (G - \langle G \rangle) |\psi\rangle, \quad \langle G \rangle = \langle \psi | G | \psi \rangle. \quad (3)$$

The pure-state QFI is

$$F_Q = 4 \langle \delta G | \delta G \rangle. \quad (4)$$

In the ratio η , the overall normalization of G cancels; the code therefore only needs a consistent G convention between numerator and denominator.

4 How F_Q is Computed

For a pure state,

$$F_Q[\psi; G] = 4 \text{Var}_{\psi}(G) = 4 (\langle G^2 \rangle - \langle G \rangle^2). \quad (5)$$

For the staggered generator

$$G = S_{\pi}^z = \frac{1}{\sqrt{N}} \sum_{j=1}^N (-1)^j S_j^z$$

used in the production observable files, this becomes

$$F_Q = 4 \left[\frac{1}{N} \sum_{j,k=1}^N (-1)^{j+k} (\langle S_j^z S_k^z \rangle - \langle S_j^z \rangle \langle S_k^z \rangle) \right]. \quad (6)$$

This is the formula used in the DMRG observable JSONs. The factor $1/\sqrt{N}$ makes G an intensive collective mode, so $F_Q = O(1)$ for the dimer and cluster product states and grows only logarithmically at the uniform Heisenberg point.

4.1 Correlation-matrix implementation

The code first computes the one-point magnetization

$$m_j^z = \langle S_j^z \rangle \quad (7)$$

and the two-point correlator

$$C_{jk}^{zz} = \langle S_j^z S_k^z \rangle. \quad (8)$$

For $j \neq k$, C_{jk}^{zz} is obtained from ITensor's `correlation_matrix`; for $j = k$ the spin-1/2 identity gives

$$\langle (S_j^z)^2 \rangle = \frac{1}{4}. \quad (9)$$

The production routine in `observables.jl` implements (6) as

```
# F_Q (probe G = N^{-1/2} sum_j (-1)^j S^z_j; F_Q = 4 Var(G)).
FQ = 0.0
for j in 1:N, k in 1:N
    sgn = ((-1)^(j+k))
    czz = j < k ? real(two_pt[3, 3, j, k]) :
           j > k ? real(two_pt[3, 3, k, j]) :
           0.25
    FQ += 4.0/N * sgn * (czz - one_pt[3, j] * one_pt[3, k])
end
```

This is equivalent to building the MPO for G , applying it to the MPS, and evaluating $4 \langle \delta G | \delta G \rangle$.

4.2 MPO implementation

The eta postprocessor uses the vector form directly. It constructs

$$G = \sum_j (-1)^j S_j^z$$

as an MPO, applies it to the MPS, and centers the result:

```
Gpsi = apply(G, psi; cutoff=1e-14)
target = Gpsi - inner(psi, Gpsi) * psi
```

Then

$$\langle \text{target} | \text{target} \rangle = \langle \delta G | \delta G \rangle \quad (10)$$

is the denominator used in η . The extra normalization by $1/\sqrt{N}$ is unnecessary inside η , because both numerator and denominator scale in the same way under $G \mapsto cG$:

$$|\delta G\rangle \mapsto c |\delta G\rangle, \quad \eta \mapsto \eta.$$

4.3 Why both F_Q methods agree

There are two equivalent computational routes:

1. compute all $S^z S^z$ connected correlations and evaluate (6);
2. build the MPS vector $|\delta G\rangle$ and evaluate $4\langle\delta G|\delta G\rangle$.

They are the same because

$$\langle\delta G|\delta G\rangle = \langle\psi|(G - \langle G\rangle)^2|\psi\rangle \quad (11)$$

$$= \langle G^2\rangle - \langle G\rangle^2 \quad (12)$$

$$= \frac{1}{N} \sum_{j,k} (-1)^{j+k} (\langle S_j^z S_k^z\rangle - \langle S_j^z\rangle\langle S_k^z\rangle). \quad (13)$$

The JSON files store the first route because it is also useful as a diagnostic of the raw spin correlations. The eta postprocessor uses the second route because it needs $|\delta G\rangle$ as an actual MPS target vector for the matched-filter projection.

5 Matched Filter on an Operator Subspace

Let

$$\mathcal{V}_{\leq k}^{\text{conn}} = \text{span}_{\mathbb{R}}\{O_a\}$$

be the real vector space spanned by Hermitian Pauli strings supported on connected windows of length at most k . For a candidate observable

$$O(\mathbf{w}) = \sum_a w_a O_a,$$

the error-propagation Fisher information is

$$F_{\text{EP}}[O(\mathbf{w})] = \frac{|-i\langle[O(\mathbf{w}), G]\rangle|^2}{\text{Var}(O(\mathbf{w}))}. \quad (14)$$

Define

$$s_a = -i\langle[O_a, G]\rangle, \quad \Sigma_{ab} = \frac{1}{2} \langle\{\delta O_a, \delta O_b\}\rangle, \quad (15)$$

where

$$\delta O_a = O_a - \langle O_a\rangle.$$

Then

$$F_{\text{EP}}[O(\mathbf{w})] = \frac{(\mathbf{w}^T \mathbf{s})^2}{\mathbf{w}^T \Sigma \mathbf{w}}. \quad (16)$$

5.1 Why the maximizer is the Wiener solution

The optimization problem is therefore

$$\max_{\mathbf{w} \neq 0} R(\mathbf{w}), \quad R(\mathbf{w}) = \frac{(\mathbf{w}^T \mathbf{s})^2}{\mathbf{w}^T \Sigma \mathbf{w}}. \quad (17)$$

This is a generalized Rayleigh quotient. The numerator is the squared linear signal extracted by the observable $O(\mathbf{w})$, while the denominator is its variance/noise. Thus the problem is exactly

the same algebraic problem as a matched filter or Wiener filter: choose weights $\mathbf{w}$ that maximize signal-to-noise.

First assume that Σ is strictly positive definite. The quotient is invariant under rescaling $\mathbf{w} \mapsto c\mathbf{w}$, so we may impose the normalization constraint

$$\mathbf{w}^T \Sigma \mathbf{w} = 1$$

and maximize

$$(\mathbf{w}^T \mathbf{s})^2.$$

Equivalently, maximize $\mathbf{w}^T \mathbf{s}$ with the same constraint; the overall sign of $\mathbf{w}$ is irrelevant because F_{EP} squares the signal. Introduce a Lagrange multiplier:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^T \mathbf{s} - \frac{\lambda}{2} (\mathbf{w}^T \Sigma \mathbf{w} - 1). \quad (18)$$

Stationarity gives

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{s} - \lambda \Sigma \mathbf{w} = 0, \quad (19)$$

so

$$\Sigma \mathbf{w} = \lambda^{-1} \mathbf{s}. \quad (20)$$

The overall scalar is immaterial, since $R(\mathbf{w})$ is scale invariant. Therefore the optimizing direction is

$$\mathbf{w}^* \propto \Sigma^{-1} \mathbf{s}. \quad (21)$$

Choosing the natural normalization gives

$$\mathbf{w}^* = \Sigma^{-1} \mathbf{s}. \quad (22)$$

Substituting back into (17),

$$R(\mathbf{w}^*) = \frac{[(\Sigma^{-1} \mathbf{s})^T \mathbf{s}]^2}{(\Sigma^{-1} \mathbf{s})^T \Sigma (\Sigma^{-1} \mathbf{s})} \quad (23)$$

$$= \frac{[\mathbf{s}^T \Sigma^{-1} \mathbf{s}]^2}{\mathbf{s}^T \Sigma^{-1} \mathbf{s}} \quad (24)$$

$$= \mathbf{s}^T \Sigma^{-1} \mathbf{s}. \quad (25)$$

Equivalently, the same result follows immediately from Cauchy–Schwarz. Write

$$\mathbf{u} = \Sigma^{1/2} \mathbf{w}, \quad \mathbf{v} = \Sigma^{-1/2} \mathbf{s}.$$

Then

$$\mathbf{w}^T \mathbf{s} = \mathbf{u}^T \mathbf{v}, \quad \mathbf{w}^T \Sigma \mathbf{w} = \mathbf{u}^T \mathbf{u}.$$

Therefore

$$R(\mathbf{w}) = \frac{(\mathbf{u}^T \mathbf{v})^2}{\mathbf{u}^T \mathbf{u}} \leq \mathbf{v}^T \mathbf{v} = \mathbf{s}^T \Sigma^{-1} \mathbf{s}, \quad (26)$$

with equality when

$$\mathbf{u} \propto \mathbf{v}, \quad \text{i.e.} \quad \Sigma^{1/2} \mathbf{w} \propto \Sigma^{-1/2} \mathbf{s},$$

again giving

$$\mathbf{w}^* \propto \Sigma^{-1} \mathbf{s}.$$

5.2 What Σ^+ means

In practice, Σ need not be strictly invertible. This is not a numerical accident: the operator list often contains redundant directions. For example, two different Pauli strings may create the same vector after acting on $|\psi\rangle$, or a string may annihilate the state entirely. Such directions have zero variance and therefore lie in the null space of Σ .

The symbol

$$\Sigma^+$$

means the Moore–Penrose pseudoinverse. If

$$\Sigma = V \Lambda V^T = \sum_{\alpha} \lambda_{\alpha} \mathbf{v}_{\alpha} \mathbf{v}_{\alpha}^T \quad (27)$$

with $\lambda_{\alpha} \geq 0$, then

$$\Sigma^+ = \sum_{\alpha: \lambda_{\alpha} > 0} \lambda_{\alpha}^{-1} \mathbf{v}_{\alpha} \mathbf{v}_{\alpha}^T. \quad (28)$$

Zero-variance directions are discarded rather than inverted.

With a singular covariance matrix, decompose the operator coefficient vector into range and null components,

$$\mathbf{w} = \mathbf{w}_{\text{ran}} + \mathbf{w}_{\text{null}}, \quad \Sigma \mathbf{w}_{\text{null}} = 0.$$

The variance depends only on $\mathbf{w}_{\text{ran}}$. A physically sensible observable cannot obtain finite signal from a strictly zero-variance direction; equivalently, the signal vector $\mathbf{s}$ lies in the range of Σ for exact arithmetic. Numerically, we project onto the positive-eigenvalue subspace. The optimizing minimum-norm coefficient vector is then

$$\mathbf{w}^{\star} = \Sigma^+ \mathbf{s}, \quad (29)$$

and the optimal extracted Fisher information is

$$\eta_{\mathcal{V}}^{\star} = \frac{\mathbf{s}^T \Sigma^+ \mathbf{s}}{F_Q}. \quad (30)$$

This is what the code implements. In the notation of the MPS Gram matrix below, Σ is represented by the real symmetric Gram matrix K , and K^+ is applied by diagonalizing K , discarding eigenvalues below a tolerance, and dividing only by the retained eigenvalues.

6 Equivalent Projection Formula

For implementation, it is more stable to use the Hilbert-space projection form. Define

$$|\phi_a\rangle = \delta O_a |\psi\rangle = (O_a - \langle O_a \rangle) |\psi\rangle. \quad (31)$$

Then

$$\eta_{\mathcal{V}}^{\star} = \frac{\|P_{\text{span}_{\mathbb{R}}\{|\phi_a\rangle\}} |\delta G\rangle\|^2}{\langle \delta G | \delta G \rangle}. \quad (32)$$

For a real Hermitian operator subspace, the ED code uses the real-form embedding

$$\Phi_{\text{real}} = \begin{bmatrix} \text{Re } \phi_1 & \cdots & \text{Re } \phi_M \\ \text{Im } \phi_1 & \cdots & \text{Im } \phi_M \end{bmatrix}, \quad t = \begin{bmatrix} \text{Im } \delta G \\ -\text{Re } \delta G \end{bmatrix}. \quad (33)$$

Then

$$\eta_{\mathcal{V}}^* = \frac{\|P_{\text{col}(\Phi_{\text{real}})}t\|^2}{\|t\|^2}. \quad (34)$$

The MPS code computes this same projection without expanding the full 2^N vector.

7 Symmetry Reduction: Odd- Y Sector

The DMRG states in this campaign are real and live in the $S_{\text{tot}}^z = 0$ sector. The generator $G = S_{\pi}^z$ is real and diagonal, so

$$|\delta G\rangle$$

is a real vector. In the real-form target

$$t = \begin{bmatrix} \text{Im } \delta G \\ -\text{Re } \delta G \end{bmatrix},$$

the first block vanishes and the second block is nonzero.

A Hermitian Pauli string with an even number of Y operators maps a real state to a real vector. Its real-form column has support in the first block:

$$\begin{bmatrix} \text{Re } \phi_a \\ 0 \end{bmatrix}.$$

It therefore has zero direct overlap with t .

A Hermitian Pauli string with an odd number of Y operators maps a real state to an imaginary vector. Its real-form column has support in the second block:

$$\begin{bmatrix} 0 \\ \text{Im } \phi_a \end{bmatrix},$$

and can carry matched-filter signal.

The second exact filter comes from the unbroken $U(1)$ spin symmetry. The DMRG state lies in the $S_{\text{tot}}^z = 0$ sector, and $G = S_{\pi}^z$ commutes with S_{tot}^z . Therefore

$$|\delta G\rangle$$

also lies in the $S_{\text{tot}}^z = 0$ sector. Any operator string whose ladder-operator decomposition has no $\Delta S^z = 0$ component creates a state orthogonal to $|\delta G\rangle$ and cannot contribute to the matched filter.

In the Cartesian Pauli basis, each S^x or S^y contributes one transverse ladder operator:

$$S^x = \frac{1}{2}(S^+ + S^-), \quad S^y = \frac{1}{2i}(S^+ - S^-).$$

A product with an odd number of transverse factors has only odd total ΔS^z sectors and therefore no $\Delta S^z = 0$ component. Hence we must also require

$$\#\{i : c_i \in \{1, 2\}\} \equiv 0 \pmod{2}.$$

Therefore, for this real-state, S_{π}^z problem, the code enumerates only connected Pauli strings satisfying both

$$\#Y \equiv 1 \pmod{2}, \quad \#(X, Y) \equiv 0 \pmod{2}.$$

The first condition is the real-form/time-reversal filter; the second is the $U(1)$ spin-charge filter. Both are exact for this campaign. As a sanity check, on the dimer state the reduced basis gives

$$\eta_{\leq 2} = 1, \quad \eta_{\leq 4} = 1,$$

matching the full ED result.

8 Operator Enumeration

The local Pauli alphabet is encoded as

$$0 = I, \quad 1 = S^x, \quad 2 = S^y, \quad 3 = S^z.$$

For a window of length w , a code tuple

$$c = (c_1, \dots, c_w)$$

is included if

1. it has exact support on that window:

$$c_1 \neq 0, \quad c_w \neq 0,$$

2. and it has odd Y parity:

$$\#\{i : c_i = 2\} \equiv 1 \pmod{2}.$$

3. and it lies in the $\Delta S^z = 0$ sector:

$$\#\{i : c_i = 1 \text{ or } c_i = 2\} \equiv 0 \pmod{2}.$$

For a chain with open boundaries, windows are

$$W_{r,w} = \{r, r+1, \dots, r+w-1\}, \quad r = 1, \dots, N-w+1.$$

The cumulative basis for $\eta_{\leq k}$ is

$$\mathcal{B}_{\leq k} = \bigcup_{w=1}^k \bigcup_{r=1}^{N-w+1} \{O_{r,w,c}\}.$$

Thus $\eta_{\leq 4}$ contains all $\eta_{\leq 2}$ operators plus the additional three- and four-site connected-window strings.

9 MPS-Level Algorithm

For each saved checkpoint

$$\text{obs/T0_path/}<\text{label}>_N<N>_psi.h5,$$

the code performs the following steps.

9.1 Step 1: Load and Densify the MPS

The production DMRG used S^z quantum-number conservation. Operators containing S^x and S^y change quantum number sectors, so before applying arbitrary Pauli strings we convert the block-sparse MPS to a dense MPS:

```
f = h5open(psi_path, "r")
psi = read(f, "psi", MPS)
close(f)

psi = dense(psi)
sites = siteinds(psi)
```

At the tensor level, an open-boundary MPS is

$$|\psi\rangle = \sum_{\sigma_1, \dots, \sigma_N} A^{[1]\sigma_1} A^{[2]\sigma_2} \dots A^{[N]\sigma_N} |\sigma_1 \dots \sigma_N\rangle. \quad (35)$$

Each local tensor has one physical index σ_j and up to two bond indices:

$$A^{[j]} = A_{\alpha_{j-1}, \alpha_j}^{[j]\sigma_j}. \quad (36)$$

With quantum-number conservation turned on, the tensor is block sparse: only entries obeying the S^z flux rule are stored. A local S^x or S^y operator changes the S^z quantum number on the physical index, so applying it generally creates sectors that were absent in the block-sparse representation. The call

`dense(psi)`

removes the QN block structure and stores ordinary dense tensors, so non- S^z -conserving Pauli strings can be applied safely.

9.2 Step 2: Build the Target Vector

Construct the MPO

$$G = \sum_{j=1}^N (-1)^j S_j^z$$

and apply it to the MPS:

```
function staggered_Gpsi(psi::MPS, sites)
    N = length(psi)
    os = OpSum()
    for j in 1:N
        os += (-1)^j, "Sz", j
    end
    G = MPO(os, sites)
    out = apply(G, psi; cutoff=1e-14)
    noprime!(out)
    return out
end
```

Then center the vector:

```
function centered(phi::MPS, psi::MPS)
    ev = inner(psi, phi)
    return phi - ev * psi
end

target = centered(staggered_Gpsi(psi, sites), psi)
```

Mathematically,

$$|\text{target}\rangle = |\delta G\rangle = (G - \langle G\rangle) |\psi\rangle.$$

9.3 Step 3: Apply Each Pauli String

For each connected-window Pauli string O_a , apply the local operators directly to a copy of the MPS:

```
const PAULI_OPS = Dict{1 => "Sx", 2 => "Sy", 3 => "Sz"}

function apply_pauli_string(psi::MPS, sites, start::Int, codes::Vector{Int})
    phi = copy(psi)
    for (off, code) in enumerate(codes)
        code == 0 && continue
        j = start + off - 1
        Oj = dense(op(PAULI_OPS[code], sites[j]))
        phi[j] = Oj * phi[j]
        noprime!(phi[j])
    end
    return phi
end
```

Then center it:

```
phi_a = centered(apply_pauli_string(psi, sites, start, codes), psi)
```

This gives

$$|\phi_a\rangle = (O_a - \langle O_a \rangle) |\psi\rangle.$$

9.4 What local operator application means

Suppose a one-site operator O_j has matrix elements

$$(O_j)_{\tau_j \sigma_j}$$

on the physical spin-1/2 basis. Applying it to the MPS changes only the local tensor at site j :

$$\tilde{A}_{\alpha_{j-1}, \alpha_j}^{[j] \tau_j} = \sum_{\sigma_j} (O_j)_{\tau_j \sigma_j} A_{\alpha_{j-1}, \alpha_j}^{[j] \sigma_j}. \quad (37)$$

All other tensors are unchanged. A Pauli string on a connected window

$$O_a = P_r^{(c_1)} P_{r+1}^{(c_2)} \dots P_{r+w-1}^{(c_w)}$$

is applied by repeating (37) on each site in the window. This is precisely what the code

```
phi[j] = Oj * phi[j]
noprime!(phi[j])
```

does. In ITensor notation, O_j carries a primed output site index and an unprimed input site index. Multiplying it into `phi[j]` contracts the input physical leg with the tensor's site index and leaves a primed output site index. The call `noprime!` renames that output leg back to the standard site index so the resulting object is again a valid MPS with the same site index set as $|\psi\rangle$.

No global Hilbert-space vector is formed. The cost of applying a length- w Pauli string is just w local tensor contractions, followed by MPS inner products when that vector is used in the Gram matrix.

9.5 Step 4: Build the Gram Matrix and Target Vector

The code never forms a 2^N Hilbert-space vector. Instead it evaluates all needed quantities with MPS inner products:

```
K[a,c] = real(inner(phis[a], phis[c]))
b[a] = imag(inner(phis[a], target))
```

The real symmetric matrix

$$K_{ab} = \text{Re} \langle \phi_a | \phi_b \rangle$$

is the covariance/Gram matrix of the real operator-action subspace. The vector

$$b_a = \text{Im} \langle \phi_a | \delta G \rangle$$

is the real-form target overlap. Up to the common normalization conventions, this is the matched-filter signal s_a .

9.6 How MPS inner products are evaluated

For two MPSs

$$|\phi\rangle = \sum_{\sigma} B^{[1]\sigma_1} \dots B^{[N]\sigma_N} |\sigma\rangle, \quad |\chi\rangle = \sum_{\sigma} C^{[1]\sigma_1} \dots C^{[N]\sigma_N} |\sigma\rangle,$$

the inner product is the tensor-network contraction

$$\langle \phi | \chi \rangle = \sum_{\sigma} \overline{B^{[1]\sigma_1} \dots B^{[N]\sigma_N}} C^{[1]\sigma_1} \dots C^{[N]\sigma_N}. \quad (38)$$

ITensor evaluates this from left to right by building an environment:

$$L_j = \sum_{\sigma_1, \dots, \sigma_j} \overline{B^{[1]\sigma_1} \dots B^{[j]\sigma_j}} C^{[1]\sigma_1} \dots C^{[j]\sigma_j}. \quad (39)$$

This avoids ever expanding the 2^N coefficient vector. The final scalar is what `inner(phi, chi)` returns.

Thus the Gram element

$$K_{ab} = \text{Re} \langle \phi_a | \phi_b \rangle$$

is computed by contracting two MPSs produced by local Pauli-string actions. Similarly,

$$b_a = \text{Im} \langle \phi_a | \delta G \rangle$$

is computed by contracting the Pauli-string-action MPS with the MPO application MPS $|\delta G\rangle$.

9.7 Step 5: Solve the Projection / Wiener Problem

Diagonalize

$$K = V \Lambda V^T.$$

Discard eigenvalues below a numerical tolerance

$$\lambda_{\alpha} \leq \epsilon_{\text{rel}} \max(1, \lambda_{\text{max}}).$$

Then compute

$$\text{num} = \sum_{\alpha: \lambda_{\alpha} > 0} \frac{[(V^T b)_{\alpha}]^2}{\lambda_{\alpha}}. \quad (40)$$

Finally

$$\eta_{\leq k} = \frac{\text{num}}{\langle \delta G | \delta G \rangle}. \quad (41)$$

The corresponding code is:

```
function eta_from_basis(phis::Vector{MPS}, target::MPS; eig_rtol::Float64=1e-10)
    M = length(phis)
    K = zeros(Float64, M, M)
    b = zeros(Float64, M)

    for a in 1:M
        b[a] = imag(inner(phis[a], target))
        for c in 1:a
            val = real(inner(phis[a], phis[c]))
            K[a, c] = val
            K[c, a] = val
        end
    end

    K = 0.5 .* (K .+ K')
    evals, vecs = eigen(Symmetric(K))

    maxeval = maximum(abs.(evals))
    tol = max(1.0, maxeval) * eig_rtol
    keep = findall(>(tol), evals)

    coeff = vecs[:, keep]' * b
    num = sum(abs2.(coeff ./ sqrt.(evals[keep])))
    norm_target = real(inner(target, target))

    return num / norm_target, length(keep)
end
```

10 Cumulative Computation of $\eta_{\leq 2}$ and $\eta_{\leq 4}$

The implementation accumulates the basis by increasing window size. For each k , all strings of length k are appended to the existing basis, and the projection is recomputed:

```
phis = MPS[]

for k in 1:kmax
    for start in 1:(length(psi) - k + 1)
        for codes in odd_y_codes(k)
            phi = centered(apply_pauli_string(psi, sites, start, codes), psi)
            push!(phis, phi)
        end
    end
end
```

```

eta, rank = eta_from_basis(phis, target; eig_rtol=eig_rtol)
eta_by_k[string(k)] = eta
rank_by_k[string(k)] = rank
nops_by_k[string(k)] = length(phis)
end

```

Thus the reported $k = 4$ value is truly

$$\eta_{\leq 4},$$

not the contribution of exactly four-site windows alone.

11 Submitted Production Job

The current production postprocessing is submitted as

```
qfi_eta_path_T0.slurm.
```

It runs one path checkpoint per array task:

```

julia --project="$QFI_JULIA_PROJECT" compute_eta_path_T0.jl \
  --input-file "$IN" \
  --out "$OUT" \
  --kmax 4 \
  --eig-rtol 1e-10

```

The output for each state is

```
obs/T0_path_eta/<label>_N<N>_eta.json.
```

Each JSON contains

- the path label and couplings (J_1, J_2, J_3) ,
- N , E_0 , χ_{final} , and F_Q ,
- $\eta_{\leq 1}, \eta_{\leq 2}, \eta_{\leq 3}, \eta_{\leq 4}$,
- the numerical rank retained in each Gram solve,
- the number of operator-action vectors included.

12 Sanity Check

Before launching the full path job, the code was tested on the corrected dimer state $D = (1, 0, 0)$ at $N = 16$. Since this is a product of isolated singlets, the matched filter must saturate already at $k = 2$:

$$\eta_{\leq 2}(D) = 1.$$

The MPS postprocessor returned

$$\eta_{\leq 2} = 1.00000000000000016, \quad \eta_{\leq 4} = 1.00000000000000007.$$

This confirms the operator basis, real-form projection, and MPS inner product implementation are consistent with the ED matched-filter definition.

13 Final Figure to Produce

After all 39 postprocessing tasks complete, the final important figure is

$$\eta_{\leq 2}, \eta_{\leq 4} \quad \text{along} \quad D \rightarrow C \rightarrow U \rightarrow D$$

for

$$N = 16, 32, 64.$$

Together with the already-generated F_Q path figure, this gives the two central numerical diagnostics for the chosen generator and chosen observable algebra:

$$\boxed{F_Q \text{ and } \eta_{\leq k}.$$

The interpretation is:

- F_Q gives the total staggered response strength.
- $\eta_{\leq k}$ gives the fraction of that response visible to connected local windows of size at most k .

Therefore the pair

$$(F_Q, \eta_{\leq k})$$

separates “how much response exists” from “how that response can be read out.” Along the corrected $D \rightarrow C \rightarrow U \rightarrow D$ path, the expectation is:

- at the dimer point D , the response is visible already through two-site bond operators;
- at the four-site cluster point C , the response should be visible through four-site windows;
- at the uniform Heisenberg point U , fixed finite windows should capture a decreasing fraction as N grows, because the response is scale-delocalized.

This statement is intentionally about the *operator channel* of the response. In another phase, such as a Haldane SPT phase, the correct operator channel may be string-like rather than local-window-like. In that case one should compare

$$\eta_{\text{local}}(k) \quad \text{against} \quad \eta_{\text{string}}(k, \ell),$$

rather than interpreting a small local-window value as long-range entanglement.